MANIPAL UNIVERSITY,JAIPUR

OPEN ELECTIVE

INTRODUCTION TO INTELLIGENT SYSTEM

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SECTION: G

Title:

Design of Fuzzy Logic System and Hybrid Intelligent System using MATLAB

Introduction

Fuzzy logic is a computational approach that deals with uncertainty and imprecision, similar to human decision-making. It is widely used in control systems such as washing machines, air conditioners, and traffic systems.

In this project, two systems are developed:

- **A fuzzy logic system to determine washing machine cycle time.**
- **A hybrid intelligent system combining fuzzy logic and neural networks (ANFIS) to predict student performance.**

Q1: Fuzzy Logic System for Washing Machine

Objective

To design a fuzzy logic system that determines washing machine cycle time based on:

- **Dirt Level**
- **Load Size**

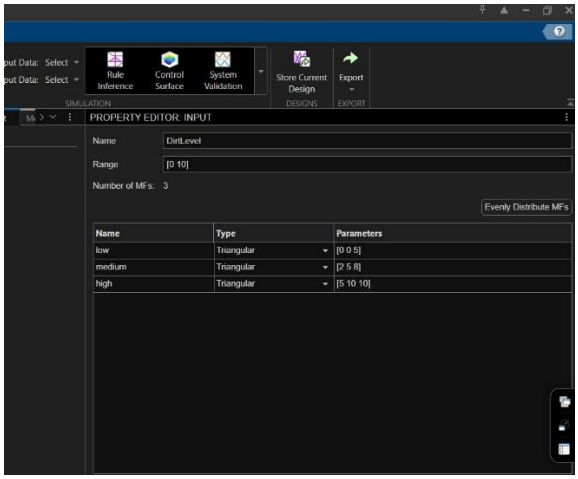
Input and Output Variables

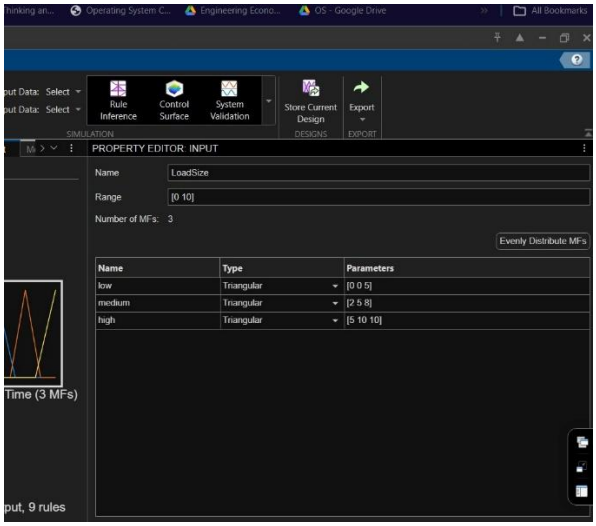
Inputs:

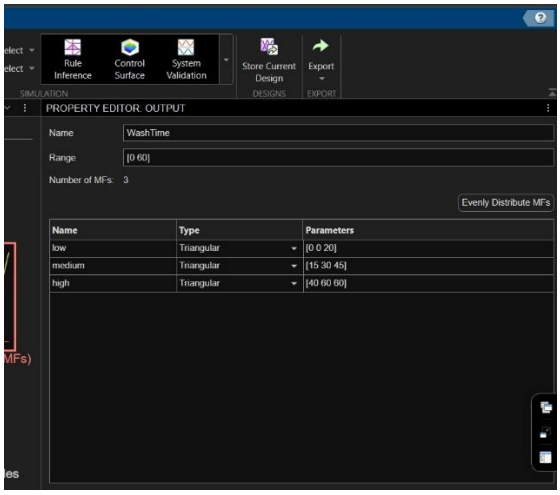
1. **Dirt Level (Range: 0–10)**
    - **Low**
    - **Medium**
    - **High**
2. **Load Size (Range: 0–10)**
    - **Low**
    - **Medium**
    - **High**

Output:

1. **Wash Time (Range: 0–60 minutes)**
    - **Low**
    - **Medium**
    - **High**







Membership Functions

Triangular membership functions were used for all variables.

Dirt Level:

- **Low → [0 0 5]**
- **Medium → [2 5 8]**
- **High → [5 10 10]**

Load Size:

- **Low → [0 0 5]**
- **Medium → [2 5 8]**
- **High → [5 10 10]**

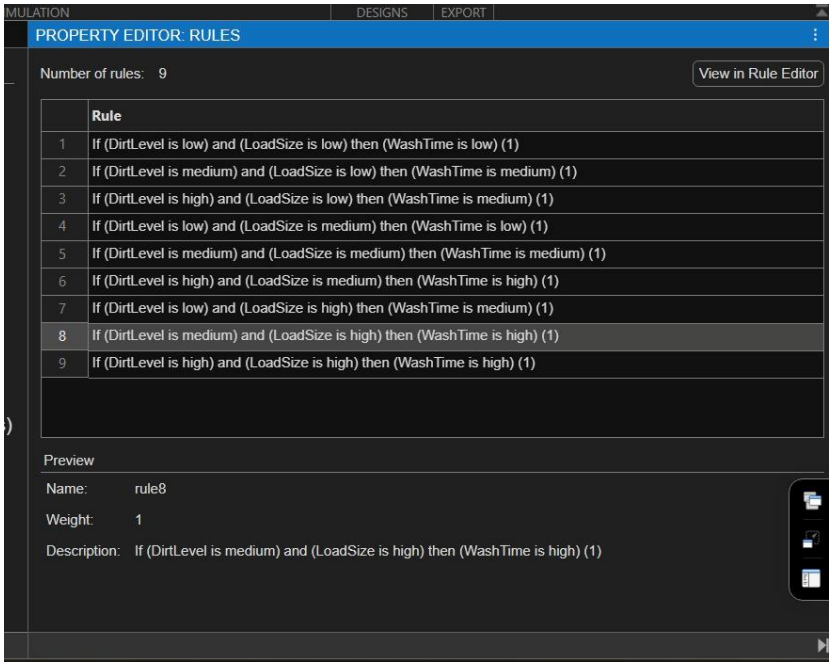
Wash Time:

- **Low→ [0 0 20]**
- **Medium → [15 30 45]**
- **High→ [40 60 60]**

Fuzzy Rules

A total of 9 fuzzy rules were created:

1. **If DirtLevel is Low and LoadSize is Low then WashTime is Low**
2. **If DirtLevel is Medium and LoadSize is Low then WashTime is Medium**
3. **If DirtLevel is High and LoadSize is Low then WashTime is Medium**
4. **If DirtLevel is Low and LoadSize is Medium then WashTime is Low**
5. **If DirtLevel is Medium and LoadSize is Medium then WashTime is Medium**
6. **If DirtLevel is High and LoadSize is Medium then WashTime is High**
7. **If DirtLevel is Low and LoadSize is High then WashTime is Medium**
8. **If DirtLevel is Medium and LoadSize is High then WashTime is High**
9. **If DirtLevel is High and LoadSize is High then WashTime is High**

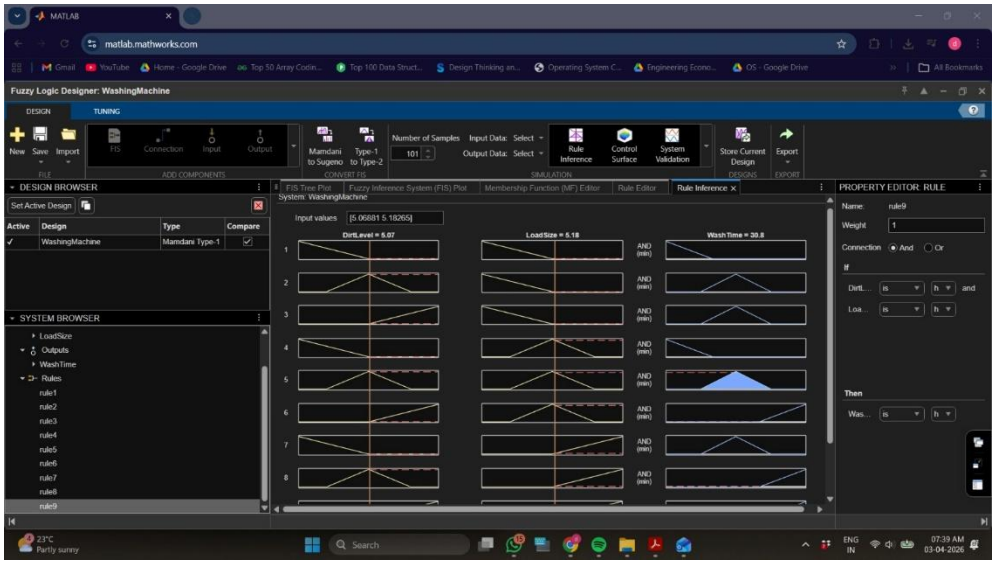


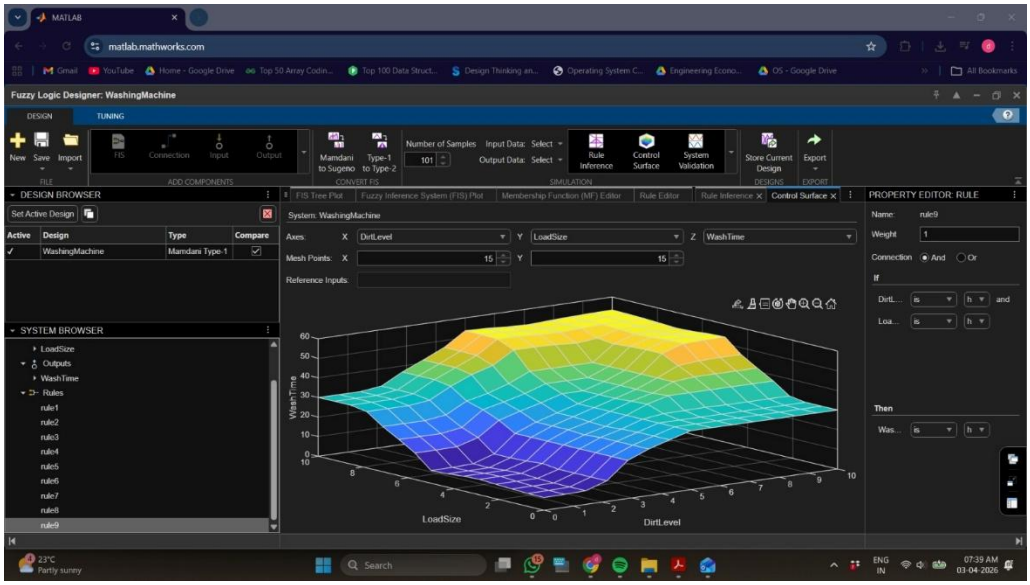
Results

The fuzzy system was tested using the Rule Viewer.

Example:

- **Input: DirtLevel = 5, LoadSize = 5**
- **Output: WashTime ≈ 30 minutes**





Q2: Hybrid Intelligent System (ANFIS)

Objective

To develop a hybrid system combining fuzzy logic and neural networks to predict student performance.

Input and Output Variables

Inputs:

- **Attendance**
- **Assignment Marks**
- **Test Marks**

Output:

- **Performance Level**

Dataset

Sample dataset used:

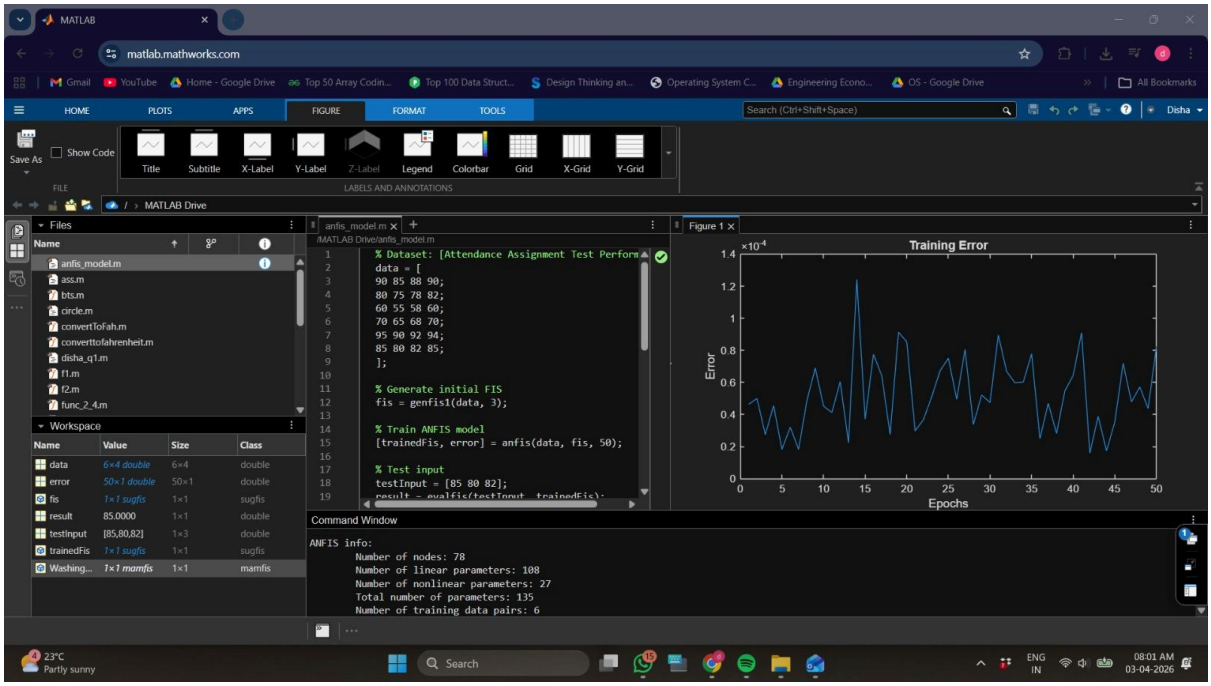
Attendance	Assignment	Test	Performance
90	85	88	90
80	75	78	82
60	55	58	60
70	65	68	70
95	90	92	94
85	80	82	85

ANFIS Implementation

The ANFIS model was implemented using MATLAB.

Steps:

1. **Dataset was defined**
2. **Initial FIS was generated using genfis1()**
3. **Model was trained using anfis()**
4. **Output was evaluated using evalfis()**

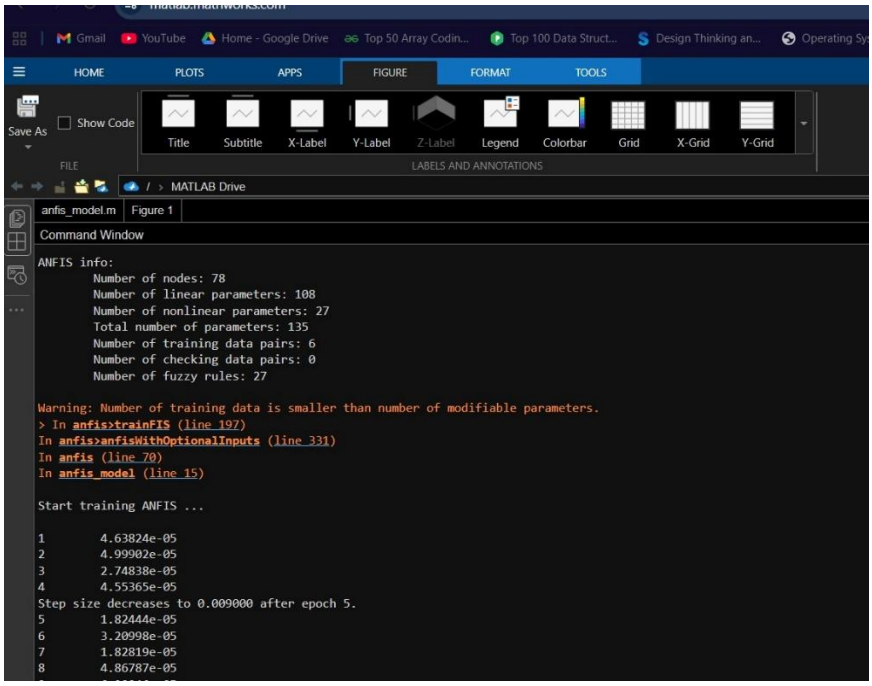


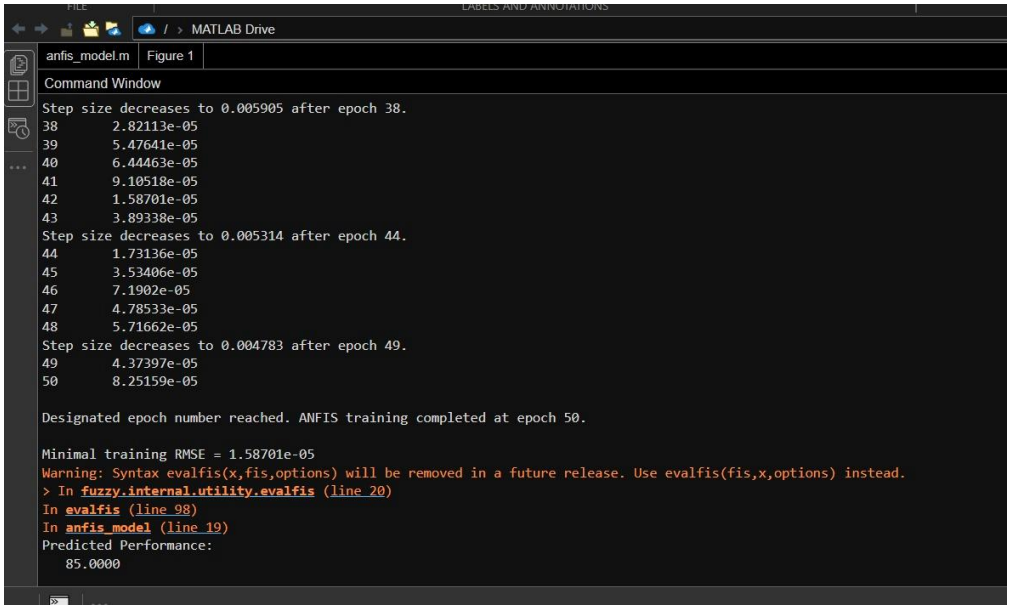
Results

- **The model successfully predicted student performance**
- **Training error graph was generated**

Performance Classification

- **0–50 → Poor**
- **50–75 → Average**
- **75–100 → Good**





Integration of Fuzzy Logic and Neural Network

The hybrid system combines:

- **Fuzzy Logic: Defines rules and membership functions**
- **Neural Network: Learns from data**

The neural network adjusts the membership functions automatically to improve prediction accuracy.

Conclusion

The fuzzy logic system successfully determines washing machine cycle time based on input conditions. The hybrid ANFIS model improves prediction accuracy for student performance by combining fuzzy logic with machine learning techniques.

This project demonstrates the effectiveness of intelligent systems in solving real-world problems.